

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Founded by Richard C. Cabot
 Eric S. Rosenberg, M.D., *Editor*
 David M. Dudzinski, M.D., Meridale V. Baggett, M.D., Kathy M. Tran, M.D.,
 Dennis C. Sgroi, M.D., Jo-Anne O. Shepard, M.D., *Associate Editors*
 Emily K. McDonald, Tara Corpuz, *Production Editors*



Case 29-2021: A 12-Month-Old Boy with Fever and Developmental Regression

Madeleine I. Matthiesen, M.D., Randheer Shailam, M.D.,
 and Holly Rothermel, M.D.

PRESENTATION OF CASE

From the Departments of Medicine (M.I.M.), Pediatrics (M.I.M., H.R.), and Radiology (R.S.), Massachusetts General Hospital, and the Departments of Medicine (M.I.M.), Pediatrics (M.I.M., H.R.), and Radiology (R.S.), Harvard Medical School — both in Boston.

N Engl J Med 2021;385:1220-9.
 DOI: 10.1056/NEJMcpc2107346
 Copyright © 2021 Massachusetts Medical Society.

CME
 at NEJM.org

Dr. Emma C. Materne (Pediatrics): A 12-month-old boy with Hirschsprung's disease was admitted to this hospital because of fever.

Fifteen days before the current admission, the patient had a temperature of 39.7°C. His parents noticed that he was fussy and less active than usual and that he had a decreased appetite along with several episodes of "spitting up"; there was no vomiting or diarrhea. During the next 5 days, fever occurred daily and abated with the administration of acetaminophen or ibuprofen. Ten days before the current admission, a bowel movement with watery, nonbloody, foul-smelling, mustard-yellow stool occurred. The patient was brought to the emergency department of this hospital.

On examination, the patient was irritable. The abdomen was soft and nontender. There were well-circumscribed, erythematous, edematous plaques on the abdomen, legs, feet, and forearms. The alanine aminotransferase level was 131 U per liter (reference range, 10 to 55), the aspartate aminotransferase level 250 U per liter (reference range, 9 to 80), the ferritin level 3826 µg per liter (reference range, 30 to 300), and the D-dimer level 3218 ng per milliliter (reference range, <500). The erythrocyte sedimentation rate was 40 mm per hour (reference range, 0 to 15); the white-cell count was normal. Other laboratory test results are shown in Table 1. A radiograph of the abdomen showed gas-filled loops of bowel. The rectum was irrigated, and intravenous piperacillin-tazobactam was administered. On the third hospital day, the patient was afebrile and able to take formula, and the diarrhea abated. He was discharged home with a prescription for a course of metronidazole.

During the next 2 days, the patient had no fever or diarrhea. However, 4 days before the current admission, a day-care worker observed that the patient appeared tired. During the subsequent 3 days at home, fever again developed, with chills and a decreased appetite. The day before the current admission, the patient cried through the night. He was brought to the emergency department of this hospital for evaluation.

Table 1. Laboratory Data.

Variable	Reference Range*	On Arrival, First Admission	On Arrival, Current Admission
White-cell count (per μ l)	6000–17,500	8180	26,820
Differential count (per μ l)			
Neutrophils	1500–8600	3590	20,060
Lymphocytes	360–11,700	3980	4910
Monocytes	100–1900	530	460
Eosinophils	0–1400	20	1150
Basophils	0–500	30	0
Platelet count (per μ l)	150,000–400,000	388,000	682,000
Hemoglobin (g/dl)	10.5–13.5	10.4	10.3
Hematocrit (%)	33.0–39.0	34.4	33.8
Prothrombin time (sec)	11.5–14.5	13.1	13.8
Prothrombin-time international normalized ratio	0.9–1.1	1.0	1.1
D-dimer (ng/ml)	<500	3218	2389
Fibrinogen (mg/dl)	150–400	290	469
Ferritin (μ g/liter)	30–300	3826	689
Erythrocyte sedimentation rate (mm/hr)	0–15	40	75
C-reactive protein (mg/liter)	<8.0	32.4	114.7
Procalcitonin (ng/ml)	0.00–0.08	—	0.59
Creatinine (mg/dl)†	0.30–1.00	0.22	0.21
Alanine aminotransferase (U/liter)	10–55	131	40
Aspartate aminotransferase (U/liter)	9–80	250	50
Alkaline phosphatase (U/liter)	142–335	163	176
Lactate dehydrogenase (U/liter)	13–60	498	490

* Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

† To convert the values for creatinine to micromoles per liter, multiply by 88.4.

The parents reported that the patient was fussy and had poor sleep and generalized weakness; there was no diarrhea or new rash. The parents did not think that the patient was in pain. Additional history was obtained from the parents. The patient had met developmental milestones until 1 month earlier, when he had stopped crawling. He had previously been able to pull his body into a standing position, but he no longer attempted this maneuver. If he was placed in a standing position, he collapsed; if he was placed in a sitting position, he was able to stay upright. He had previously passed objects from hand to hand, but he had a new preference of holding objects with the left hand. Two days

before this evaluation, the parents had noticed that the patient no longer walked with support; he seemed to dislike having his hands held.

The patient had cornea plana and a history of Hirschsprung's disease, for which he had undergone laparoscopic sigmoid colectomy and an anal pull-through procedure at 2 months of age. Four months before the current evaluation, an intermittent diffuse rash on the trunk, arms, and legs had developed. The rash was characterized by purplish and pinkish plaques that appeared and varied in intensity of color throughout the day; it did not seem to be itchy or painful and usually resolved within 24 hours without treatment.

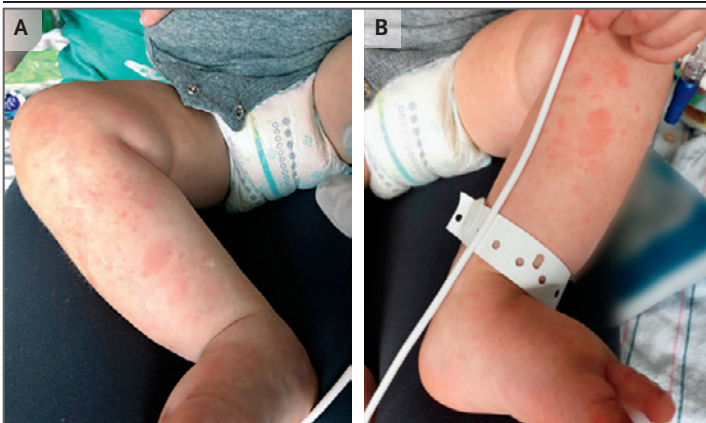


Figure 1. Clinical Photographs.

The patient has well-circumscribed, erythematous, edematous plaques involving the back, abdomen, arms, and legs (Panels A and B). The evanescent rash is nonpruritic and nontender; it does not affect the palms or soles.

Immunizations were up to date. Medications included metronidazole, acetaminophen, ibuprofen, and cetirizine, as well as carboxymethylcellulose eye drops; there were no known allergies to medications. The patient lived with his mother, father, and pet dog in a small town in New England, and he attended day care 5 days per week. His mother was healthy, and his father had irritable bowel syndrome and Poland syndrome (congenital aplasia of unilateral chest-wall muscles). His paternal grandmother had hypothyroidism, his maternal great-grandfather had had pancreatic cancer, his paternal great-grandfather had had breast cancer, a maternal second cousin had autism, and multiple maternal family members had Raynaud's phenomenon.

On examination, the temperature was 38.3°C, the blood pressure 113/52 mm Hg, the heart rate 166 beats per minute (normal range for age, 80 to 160), and the respiratory rate 30 breaths per minute (normal range for age, 30 to 60). The patient was diaphoretic and irritable, and he cried during the examination. He had rhinorrhea. The abdomen was not distended or tender. There was limited extension of the wrists and knees, as well as limited inversion and eversion of the feet. The ankles were mildly stiff, and the feet were mildly swollen. There were erythematous plaques on the abdomen, back, arms, and legs (Fig. 1).

The ferritin and D-dimer levels remained elevated, and the aspartate aminotransferase and alanine aminotransferase levels were normal. The

erythrocyte sedimentation rate was 75 mm per hour, the white-cell count 26,820 per microliter (reference range, 6000 to 17,500), and the platelet count 682,000 per microliter (reference range, 150,000 to 400,000). Blood cultures and urinalysis were normal. Testing of a nasopharyngeal swab for severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) RNA was negative. Other laboratory test results are shown in Table 1.

Dr. Randheer Shailam: A radiograph of the chest and abdomen was normal. Ultrasonography of the abdomen revealed cholelithiasis and prominent lymph nodes in the right lower quadrant. Ultrasonography of the legs revealed a small amount of joint fluid in the hips and knees (Fig. 2); the amount of fluid appeared to be within the normal range. Computed tomography (CT) of the abdomen, performed after the administration of intravenous contrast material, revealed a gallstone in the neck of the gallbladder, without gallbladder distention. There were a few prominent lymph nodes in the paraaortic, iliac, and inguinal regions (Fig. 2). Transthoracic echocardiography showed normal coronary arteries.

Dr. Materne: The empirical administration of intravenous vancomycin, cefepime, and metronidazole was begun. During the next 3 days, the fever continued and the white-cell count remained elevated. On the seventh hospital day, the fever, leukocytosis, and elevation in levels of inflammatory markers persisted. The administration of antibiotic agents was stopped.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Madeleine I. Matthiesen: I was a member of the team that cared for this patient, and I am aware of the final diagnosis. This 12-month-old boy with a history of Hirschsprung's disease was admitted to the hospital with fever.

Initially, we considered the most likely causes of this child's febrile illness. At this point, system 1 thinking — or fast thinking — was activated. System 1 thinking involves intuitive and efficient pattern recognition, frequently relying on epidemiology and the health care provider's experience. This type of thinking allows providers to make thousands of rapid clinical decisions each day and, in this case, led to the immediate administration of antibiotics and a workup for

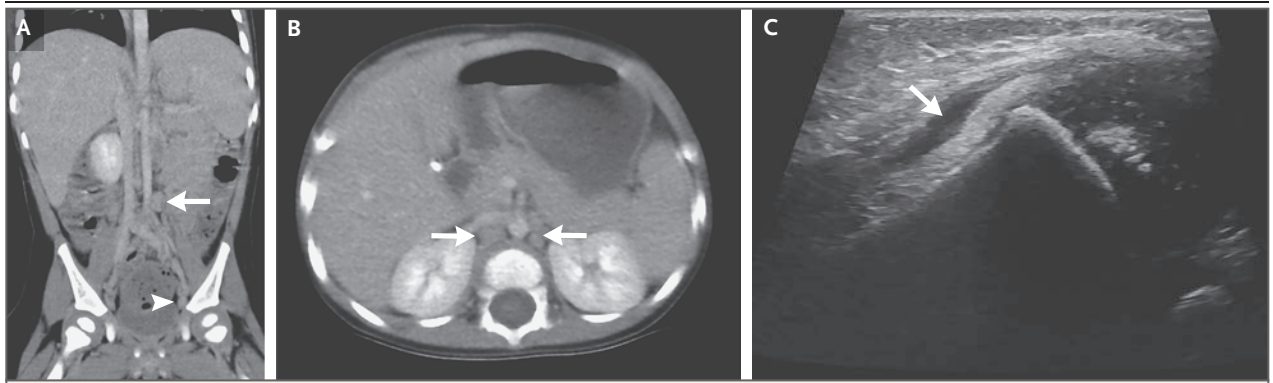


Figure 2. Imaging Studies.

Coronal and axial CT images (Panels A and B, respectively) of the abdomen and pelvis show multiple enlarged paraaortic lymph nodes (arrows) and an enlarged iliac lymph node (arrowhead). The liver and spleen have a normal appearance. There is no evidence of cholecystitis, although a single gallstone was detected (not shown). There is no evidence of cancer. A sagittal ultrasound image (Panel C) shows joint fluid in the right knee (arrow); small effusions involving both knees and hips were detected (not shown).

coronavirus disease 2019 (Covid-19) and related illnesses.

One problem with system 1 thinking is the difficulty distinguishing appropriate from inappropriate heuristic activation. A heuristic is an automatically and unconsciously deployed mental shortcut that can facilitate rapid decision making. Heuristics have the potential to be based on invalid relationships and thus to represent cognitive bias.¹ To avoid cognitive bias, deliberate reflection is critical.

Thus, we used deliberate reflection to consider the aspects of the case that did and did not fit the given diagnoses; this process revealed system 1 thinking to be incorrect. At this point, system 2 thinking — or slow thinking — was activated. Although it is time-intensive, system 2 thinking is an important tool for consideration of a broad differential diagnosis in the context of diagnostic uncertainty and, in this case, led to further exploration of the patient's developmental regression, lymphadenopathy, and rash in the presence of persistent fever that did not abate despite the administration of antibiotics.

HIRSCHSPRUNG-ASSOCIATED ENTEROCOLITIS

Hirschsprung-associated enterocolitis is the most common reason for hospitalization among patients with Hirschsprung's disease; it occurs in approximately one third of patients with Hirschsprung's disease.² Because epidemiologic associations are a key contributor to system 1 thinking,

this diagnosis was considered and treatment was initiated immediately.

Although this patient had undergone sigmoid colectomy, some risk for Hirschsprung-associated enterocolitis was still present postoperatively because of bacterial translocation.³ The symptoms that had resulted in his first hospitalization, including abdominal distention, diarrhea, and fever, were all consistent with this diagnosis.

Deliberate reflection on this system 1 diagnosis revealed inappropriate anchoring on the diagnosis established during the previous hospitalization. The description of the patient's stool, the resolution of abdominal distention before fever recrudescence, and the ongoing fever despite the multiple-day course of antibiotics all made the diagnosis of Hirschsprung-associated enterocolitis unlikely.

MULTISYSTEM INFLAMMATORY SYNDROME IN CHILDREN

System 1 thinking also led us to consider Covid-19 and related illnesses, since the patient presented with fever and evidence of systemic inflammation during the pandemic. Although Covid-19 was unlikely because of the low community positivity rates at the time and negative testing for SARS-CoV-2, the patient was hospitalized in part out of concern about multisystem inflammatory syndrome in children (MIS-C).

An availability heuristic is a cognitive bias that in this case connected the patient's presentation to examples of MIS-C that came easily to

mind in the weeks after the first wave of Covid-19 in this region. However, the estimated incidence of MIS-C is 2 to 5 per 1 million children,⁴⁻⁶ and on deliberate reflection, the patient did not meet the criteria for MIS-C outlined by the Centers for Disease Control and Prevention⁷; he had neither recent evidence of Covid-19 nor involvement of multiple (at least two) organ systems, with some combination of active cardiac, respiratory, hematologic, gastrointestinal, dermatologic, or neurologic symptoms at presentation.

FEVER OF UNKNOWN ORIGIN

Because system 1 thinking did not lead to identification of the diagnosis, system 2 thinking was activated, allowing for a systematic approach to the differential diagnosis with the use of a diagnostic schema. Schemas require a specific clinical finding to be used as a hook, and it was critical to determine the features of this case that represented the “signal” in the foreground of the illness, rather than the “noise” in the background.

Three days into the patient’s hospitalization, he was considered to have fever of unknown origin, given that the initial treatment and workup had neither elucidated the diagnosis nor led to clinical improvement. Fever of unknown origin was used as the first hook. It was undoubtedly a signal, because it had to be explained by the final diagnosis. It was also chosen because it would allow us to form a broad differential diagnosis, activating a diagnostic schema that begins with a comparison of infectious and noninfectious causes.

INFECTION

In cases of fever of unknown origin, infections are most commonly identified as the cause,⁸⁻¹⁰ and this patient was initially treated for bacterial infection. However, as his hospital course progressed without improvement in his symptoms, causes of ongoing fever despite the administration of antibiotics were considered.

First, infection with inadequate control of the source could explain ongoing fever despite antibiotic therapy. An occult abscess was not detected on either abdominal CT or chest radiography, and focal bony tenderness would be expected if the patient were to have osteomyelitis.

Second, infection with inadequate antibiotic coverage due to the presence of multidrug-resis-

tant organisms or a pathogen not covered by the empirical regimen was considered. In an otherwise healthy, immunocompetent boy who was receiving broad-spectrum antibiotics and had negative blood cultures, infection with a multidrug-resistant organism was unlikely. Tickborne infections such as anaplasmosis that require treatment with doxycycline typically cause leukopenia and abnormal results on liver-function tests, findings that were not present in this case.

Third, the fever may not have been due to bacterial infection. Several viruses — including Epstein–Barr virus, cytomegalovirus, and human herpesvirus 6 — were considered, but the presence of fever, the normal results on liver-function tests, and the absence of an acute rash and oropharyngeal findings made these diagnoses unlikely. Systemic mycobacterial infection would be uncommon in a healthy boy without a history of travel and with a negative chest radiograph. Only noninfectious causes remain serious considerations in this case.

DEVELOPMENTAL REGRESSION

The time course of the appearance of symptoms may be an important factor in narrowing the differential diagnosis in this case. It was initially unclear whether the patient’s subacute developmental regression was related to his current presentation, but as the fever shifted into a subacute time frame, regression was brought to the foreground. To narrow the differential diagnosis further, the overlap of fever of unknown origin and regression of crawling was explored.

Causes of crawl regression can be grouped into mechanical causes, neuromuscular causes, and pain. Mechanical and neuromuscular causes were unlikely in this patient, given the presence of fever and the subacute time course without a focal neurologic deficit. Pain may be due to infection, cancer, and immune-mediated causes. Since infection had already been addressed, cancer and immune-mediated diagnoses were considered.

In 2 to 15% of cases of fever of unknown origin, cancer is identified as the cause,⁸⁻¹⁰ and tumors in the limbs can cause pain that leads to crawl regression. At 12 months of age, primary bone tumors are rare, but neuroblastoma is common and can cause fever, as well as crawl regression when bone metastasis is present.

However, no abdominal mass was identified on imaging studies in this case.

Nearly two thirds of patients with leukemia present with musculoskeletal symptoms due in part to overcrowding of bone marrow by cancer cells,¹¹ and leukemia is one of the most common cancers identified in patients with fever of unknown origin. However, thrombocytosis is a rare finding in pediatric patients with leukemia, and with this patient's profound leukocytosis, immature forms would be expected on a manual cell count.

In up to 25% of cases of fever of unknown origin, immune-mediated conditions are identified as the cause.⁸⁻¹⁰ To narrow the list of possible immune-mediated causes in a patient with fever of unknown origin and crawl regression due to pain, the additional systemic findings of lymphadenopathy and rash were brought to the foreground.

Kawasaki's disease is an important consideration in any child with fever and can cause lymphadenopathy, rash, and crawl regression due to arthritis. However, in patients with Kawasaki's disease, lymphadenopathy typically occurs in the cervical lymph-node chain, and the nodes tend to be larger in size than those observed in this patient. The absence of coronary changes on echocardiography in the presence of ongoing fever was also reassuring.

Systemic lupus erythematosus can also cause fever, rash, lymphadenopathy, and crawl regression. However, the rash classically associated with this illness would not disappear, and additional neurologic and renal findings and lupus-specific antibodies would be present.

SYSTEMIC JUVENILE IDIOPATHIC ARTHRITIS

Systemic juvenile idiopathic arthritis, which is a diagnosis to consider only after ruling out infection and cancer, could explain all the features of this case. The classification criteria include arthritis, fever for 2 weeks, and one of the following four symptoms: evanescent rash, generalized lymphadenopathy, hepatomegaly or splenomegaly, or serositis (Table 2).¹² Although the peak onset of systemic juvenile idiopathic arthritis occurs at 2 years of age, diagnosis in a 12-month-old child is not uncommon.¹³ The laboratory findings of thrombocytosis, leukocytosis, and elevated levels of inflammatory markers are all consistent with the diagnosis. With a condition

Table 2. Classification Criteria for Systemic Arthritis.*

Requirements

Arthritis in one or more joints

Fever for ≥ 2 weeks, with daily occurrence for ≥ 3 days

One of the following symptoms:

Evanescent (nonfixed) erythematous rash

Generalized lymphadenopathy

Hepatomegaly or splenomegaly

Serositis

Exclusions

Psoriasis or a history of psoriasis in the patient or a first-degree relative

Arthritis in an HLA-B27–positive male beginning after his sixth birthday

Ankylosing spondylitis, enthesitis-related arthritis, sacroiliitis with inflammatory bowel disease, reactive arthritis, acute anterior uveitis, or a history of one of these disorders in a first-degree relative

Presence of rheumatoid factor IgM on at least two occasions ≥ 3 months apart

* Data are from Petty et al.¹²

as rare as systemic juvenile idiopathic arthritis, which has an estimated incidence of 0.5 per 100,000 children,¹⁴ it is important to practice deliberate reflection to consider the aspects of the case that do or do not fit with the diagnosis.

The rash classically associated with systemic juvenile idiopathic arthritis evanesces with fever. In this patient, the rash was certainly evanescent in its rapid resolution, but it occurred months before fever and arthritis, which was unusual. Also, it was unclear whether the patient's lymphadenopathy was a remnant of the previous infection, rather than evidence of systemic inflammation.

Arthritis had not been confirmed in the patient. Pediatric arthritis can be challenging to diagnose because children often cannot identify stiffness or pain, instead showing a change in gait or function, as was seen in this patient.¹⁵ Arthritic joints are typically warm but not erythematous; the knee, ankle, and wrist are most commonly affected.¹⁵ Arthritis could explain this patient's crawl regression, the apparent pain when his hands were held, and the stiffness identified on examination of the joints.

The most likely diagnosis in this patient was systemic juvenile idiopathic arthritis. He met the criteria for this condition, given the presence of arthritis, fever, lymphadenopathy, and possibly rash. Because ultrasonography had been negative, magnetic resonance imaging (MRI) was

performed to obtain a more sensitive view of the joints and to rule out occult osteomyelitis and tumor. The presence of widespread lymphadenopathy on MRI would further support the diagnosis of systemic juvenile idiopathic arthritis. If arthritis were absent but more accessible lymph nodes were identified, diagnostic biopsy would have been considered.

DR. MADELEINE I. MATTHIESEN'S DIAGNOSIS

Systemic juvenile idiopathic arthritis.

ADDITIONAL IMAGING STUDIES

Dr. Shailam: On the eighth hospital day, whole-body MRI was performed (Fig. 3). Small effusions were identified in the shoulders, elbows, hips, and knees. T2-weighted imaging of the hips showed increased signal within the greater trochanter bursa of the femora. Prominent axillary, inguinal, and popliteal lymph nodes were present bilaterally.

DISCUSSION OF MANAGEMENT

Dr. Holly Rothermel: Juvenile idiopathic arthritis is a broad term that encompasses a heterogeneous group of disorders. The International League of Associations for Rheumatology (ILAR) classification system describes seven types: oligoarticular arthritis, polyarticular arthritis (rheumatoid factor–positive and rheumatoid factor–negative), enthesitis-related arthritis, psoriatic arthritis, undifferentiated arthritis, and systemic arthritis.¹² Systemic arthritis is one of the least common and most severe types. It accounts for 5 to 15% of new diagnoses of juvenile idiopathic arthritis in North America and Europe.^{16,17} Systemic arthritis most often develops in children younger than 5 years of age, although it occurs at all ages. Unlike other types of juvenile idiopathic arthritis, systemic arthritis is seen just as often in boys as in girls. Most children with other types of juvenile arthritis receive the diagnosis in the outpatient clinic, but children with systemic arthritis are quite ill and are usually evaluated in the hospital setting. Research regarding the pathogenesis of systemic juvenile idiopathic arthritis has shown a prominent role of the innate immune response, and there is a suggestion that

Figure 3 (facing page). Whole-Body MRI Scan.

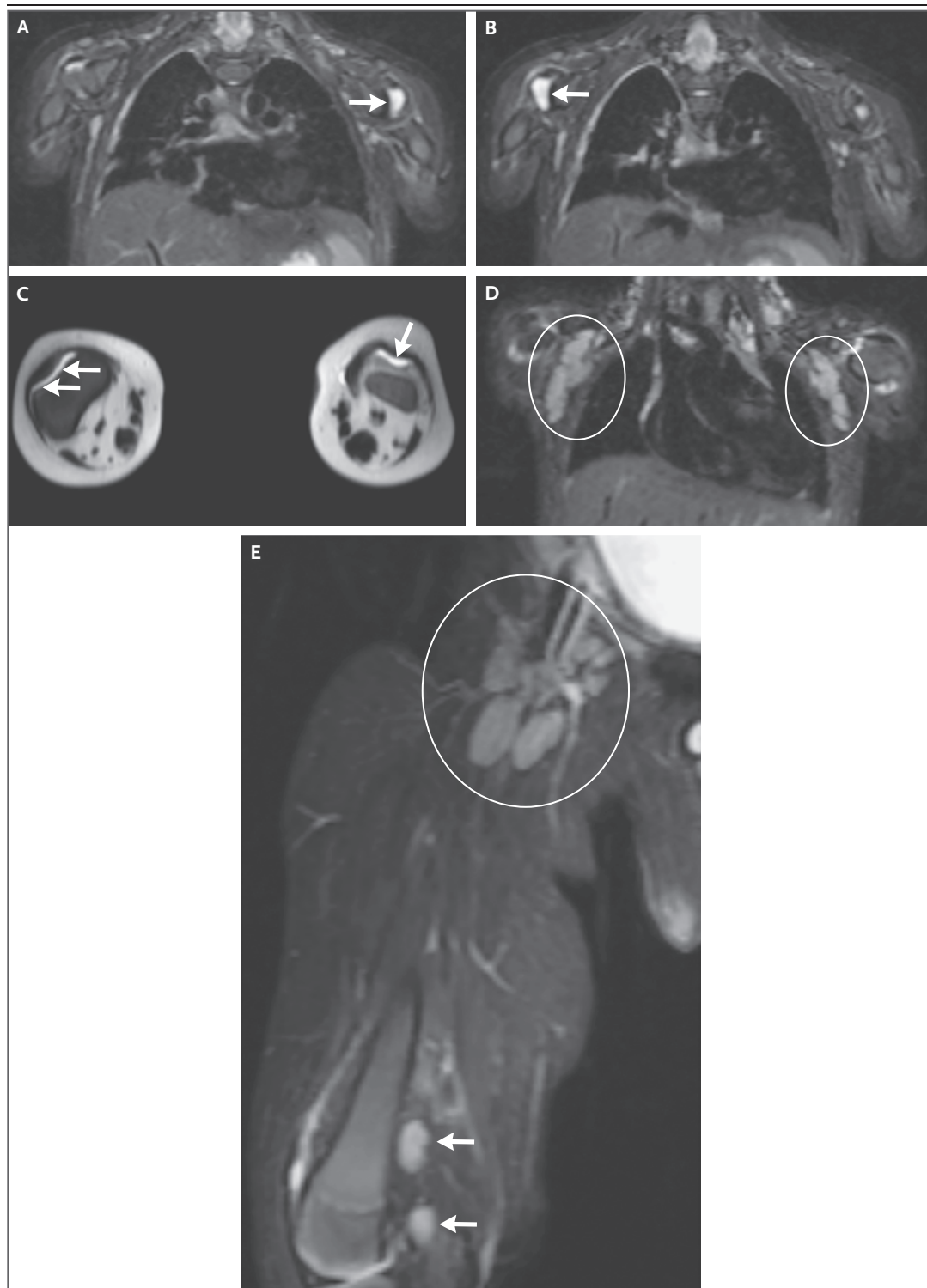
Coronal T2-weighted short-tau inversion recovery (STIR) images (Panels A and B) show small effusions involving the shoulders (arrows). An axial T2-weighted half-Fourier single-shot turbo spin–echo image (Panel C) shows small effusions involving the knees (arrows). A coronal T2-weighted STIR image of the chest (Panel D) shows prominent axillary lymph nodes bilaterally (circled). A coronal T2-weighted STIR image of the right leg (Panel E) shows prominent inguinal lymph nodes (circled) and popliteal lymph nodes (arrows); similar findings were seen on the left side (not shown).

the condition should be classified as an autoinflammatory disease rather than a classic autoimmune disease.^{18–20}

Systemic juvenile idiopathic arthritis is a clinical diagnosis that is made after all other causes have been ruled out. The fever characteristically occurs on a daily basis, with high spikes in temperature once or twice a day. The rash is described as an evanescent, salmon-colored, macular rash that is most apparent when the temperature is high. The arthritis is usually polyarticular and can involve large and small joints. Unfortunately, many patients do not meet all the ILAR diagnostic criteria. The fever pattern and rash can be atypical, and the arthritis may not appear for weeks or even months after the onset of disease.

Laboratory abnormalities associated with the illness are not specific and include leukocytosis, neutrophilia, anemia, thrombocytosis, and elevations in the erythrocyte sedimentation rate and levels of C-reactive protein, D-dimer, and ferritin. Autoantibodies such as rheumatoid factor, cyclic citrullinated peptide, antineutrophil cytoplasmic antibody, and antinuclear antibody are generally not found. Most patients who ultimately receive a diagnosis of systemic juvenile idiopathic arthritis have undergone an extensive workup to rule out alternative causes such as infection, cancer, inflammatory bowel disease, and other rheumatic diseases, including Kawasaki's disease and polyarteritis nodosa. MIS-C has now joined this list.^{12,16}

Treatment for systemic arthritis is aimed at controlling both the systemic features and the arthritis. In the past, treatment consisted of glucocorticoids, often administered at high doses, accompanied by disease-modifying antirheumatic drugs such as methotrexate. Gluco-



corticoids can lead to clinically significant toxic effects, and methotrexate does not control either the systemic features or the arthritis in many patients. With the advent of biologic agents and

continued research into the cytokine profile associated with systemic juvenile idiopathic arthritis, treatment has evolved over the past decade. The interleukin-1 inhibitors anakinra, canakinu-

mab, and rilonacept and the interleukin-6 inhibitor tocilizumab have proven very effective in treating this condition and may help to minimize or shorten the duration of glucocorticoid use.²¹⁻²⁴ Both the American College of Rheumatology and the Childhood Arthritis and Rheumatology Research Alliance have published treatment guidelines that incorporate these newer therapies.^{25,26}

Outcomes in patients with systemic juvenile idiopathic arthritis can vary. Although 40% of patients have a monophasic course with disease remission, and a small percentage of patients have a polyphasic course with periods of active disease and periods of quiescence, over half of patients have ongoing arthritis even if the systemic features have subsided.^{16,17} These patients may require further treatment with antirheumatic drugs such as methotrexate, abatacept, and tumor necrosis factor inhibitors. All patients with systemic juvenile idiopathic arthritis need to be monitored for worsening of disease and for the development of macrophage activation syndrome, a severe complication that can occur in up to 10% of patients. Macrophage activation syndrome is characterized by fever, hepatic dysfunction, purpura, and involvement of the central nervous system and can lead to multiorgan failure. Sudden decreases in the erythrocyte sedimentation rate, white-cell count, and platelet count accompanied by elevations in the levels of ferritin, liver enzymes, and triglycerides can herald this diagnosis, which must be promptly recognized and treated.²⁷ It is not yet clear whether early treatment with interleukin-1 or interleukin-6 inhibitors is associated with improved outcomes in patients with systemic arthritis, although there are some indications that it may be.²⁸

Our team met this patient and his parents a few days into his hospitalization. With the MRI finding of joint inflammation and with no clear indication of infection or cancer, we began treatment for presumed systemic juvenile idiopathic

arthritis. He started therapy with methylprednisolone and anakinra. Within 24 hours, the patient was afebrile, his rash had resolved, and he was much less fussy. He started to use his hands to grab toys and feed himself. He was discharged home 5 days later, while receiving anakinra and a tapering course of glucocorticoids.

The patient continues to be followed closely in our rheumatology clinic. By 2 months after discharge, his C-reactive protein level had decreased to 20 mg per liter (from a maximum level of 274 mg per liter) and his ferritin level to 389 μ g per liter (from a maximum level of 3826 μ g per liter). At that time, he had been weaned off glucocorticoids but was still receiving anakinra on a daily basis. His fever has not returned, but he had a brief recurrence of rash and an elevation in the C-reactive protein level; these issues resolved quickly with a weight-based dose adjustment of anakinra. He has had some evidence of persistent joint inflammation, particularly in his shoulders, for which he started treatment with methotrexate. Because of the Covid-19 pandemic, he has been unable to start in-person physical therapy, but his parents are working on his developmental skills. He resumed crawling soon after discharge and now pulls to stand, uses his hands well, and is very active. Although he responded well to anakinra therapy, the daily administration of the injections became increasingly difficult for the family and for him; we recently transitioned him to therapy with canakinumab, a longer-acting interleukin-1 inhibitor that requires only monthly dosing.

FINAL DIAGNOSIS

Systemic juvenile idiopathic arthritis.

This case was presented at the Harvard Medical School postgraduate course "Pediatric Primary Care," directed by Peter T. Greenspan, M.D., John Patrick T. Co, M.D., M.P.H., Ronni L. Goldsmith, M.D., Janice A. Lowe, M.D., and Benjamin A. Nelson, M.D.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

REFERENCES

1. Norman GR, Monteiro SD, Sherbino J, Ilgen JS, Schmidt HG, Mamede S. The causes of errors in clinical reasoning: cognitive biases, knowledge deficits, and dual process thinking. *Acad Med* 2017;92: 23-30.
2. Gosain A, Frykman PK, Cowles RA, et al. Guidelines for the diagnosis and management of Hirschsprung-associated enterocolitis. *Pediatr Surg Int* 2017;33: 517-21.
3. Gosain A, Brinkman AS. Hirschsprung's associated enterocolitis. *Curr Opin Pediatr* 2015;27:364-9.
4. Jiang L, Tang K, Levin M, et al. COVID-19 and multisystem inflammatory syndrome in children and adolescents. *Lancet Infect Dis* 2020;20(11):e276-e288.
5. Payne AB, Gilani Z, Godfred-Cato S, et al. Incidence of multisystem inflammatory syndrome in children among US per-

- sons infected with SARS-CoV-2. *JAMA Netw Open* 2021;4(6):e2116420.
6. Dufort EM, Koumans EH, Chow EJ, et al. Multisystem inflammatory syndrome in children in New York State. *N Engl J Med* 2020;383:347-58.
 7. CDC Health Alert Network. Multisystem inflammatory syndrome in children (MIS-C) associated with coronavirus disease 2019 (COVID-19). May 14, 2020 (<https://emergency.cdc.gov/han/2020/han00432.asp>).
 8. Antoon JW, Potisek NM, Lohr JA. Pediatric fever of unknown origin. *Pediatr Rev* 2015;36:380-90.
 9. Szymanski AM, Clifford H, Ronis T. Fever of unknown origin: a retrospective review of pediatric patients from an urban, tertiary care center in Washington, DC. *World J Pediatr* 2020;16:177-84.
 10. Antoon JW, Peritz DC, Parsons MR, Skinner AC, Lohr JA. Etiology and resource use of fever of unknown origin in hospitalized children. *Hosp Pediatr* 2018;8:135-40.
 11. Clarke RT, Van den Bruel A, Bankhead C, Mitchell CD, Phillips B, Thompson MJ. Clinical presentation of childhood leukaemia: a systematic review and meta-analysis. *Arch Dis Child* 2016;101:894-901.
 12. Petty RE, Southwood TR, Manners P, et al. International League of Associations for Rheumatology classification of juvenile idiopathic arthritis: second revision, Edmonton, 2001. *J Rheumatol* 2004;31:390-2.
 13. Behrens EM, Beukelman T, Gallo L, et al. Evaluation of the presentation of systemic onset juvenile rheumatoid arthritis: data from the Pennsylvania Systemic Onset Juvenile Arthritis Registry (PASOJAR). *J Rheumatol* 2008;35:343-8.
 14. Gurion R, Lehman TJA, Moorthy LN. Systemic arthritis in children: a review of clinical presentation and treatment. *Int J Inflam* 2012;2012:271569.
 15. Rosenberg AM, Oen KG. Polyarticular juvenile idiopathic arthritis. In: Petty RE, Laxer RM, Lindsley CB, Wedderburn LR, eds. *Textbook of pediatric rheumatology*. 7th ed. Philadelphia: Elsevier, 2016:217-228.e6 (<https://www.sciencedirect.com/science/article/pii/B978032324145800017X>).
 16. De Benedetti F, Schneider R. Systemic juvenile idiopathic arthritis. In: Petty RE, Laxer RM, Lindsley CB, Wedderburn LR, eds. *Textbook of pediatric rheumatology*. 7th ed. Philadelphia: Elsevier, 2016:205-216.e6 (<https://www.sciencedirect.com/science/article/pii/B9780323241458000168>).
 17. Martini A. Systemic juvenile idiopathic arthritis. *Autoimmun Rev* 2012;12:56-9.
 18. Kessel C, Hedrich CM, Foell D. Innately adaptive or truly autoimmune: is there something unique about systemic juvenile idiopathic arthritis? *Arthritis Rheumatol* 2020;72:210-9.
 19. Martini A. It is time to rethink juvenile idiopathic arthritis classification and nomenclature. *Ann Rheum Dis* 2012;71:1437-9.
 20. Vastert SJ, de Jager W, Noordman BJ, et al. Effectiveness of first-line treatment with recombinant interleukin-1 receptor antagonist in steroid-naïve patients with new-onset systemic juvenile idiopathic arthritis: results of a prospective cohort study. *Arthritis Rheumatol* 2014;66:1034-43.
 21. Giancane G, Minoia F, Davi S, Bracciolini G, Consolaro A, Ravelli A. IL-1 inhibition in systemic juvenile idiopathic arthritis. *Front Pharmacol* 2016;7:467.
 22. Nigrovic PA, Mannion M, Prince FHM, et al. Anakinra as first-line disease-modifying therapy in systemic juvenile idiopathic arthritis: report of forty-six patients from an international multicenter series. *Arthritis Rheum* 2011;63:545-55.
 23. Brunner HI, Quartier P, Alexeeva E, et al. Efficacy and safety of canakinumab in patients with systemic juvenile idiopathic arthritis with and without fever at baseline: results from an open-label, active-treatment extension study. *Arthritis Rheumatol* 2020;72:2147-58.
 24. De Benedetti F, Brunner HI, Ruperto N, et al. Randomized trial of tocilizumab in systemic juvenile idiopathic arthritis. *N Engl J Med* 2012;367:2385-95.
 25. Ringold S, Weiss PF, Beukelman T, et al. 2013 Update of the 2011 American College of Rheumatology recommendations for the treatment of juvenile idiopathic arthritis: recommendations for the medical therapy of children with systemic juvenile idiopathic arthritis and tuberculosis screening among children receiving biologic medications. *Arthritis Care Res (Hoboken)* 2013;65:1551-63.
 26. DeWitt EM, Kimura Y, Beukelman T, et al. Consensus treatment plans for new-onset systemic juvenile idiopathic arthritis. *Arthritis Care Res (Hoboken)* 2012;64:1001-10.
 27. Ravelli A, Minoia F, Davi S, et al. 2016 Classification criteria for macrophage activation syndrome complicating systemic juvenile idiopathic arthritis: a European league against rheumatism/American College of Rheumatology/Paediatric Rheumatology International Trials Organisation collaborative initiative. *Arthritis Rheumatol* 2016;68:566-76.
 28. Nigrovic PA. Review: is there a window of opportunity for treatment of systemic juvenile idiopathic arthritis? *Arthritis Rheumatol* 2014;66:1405-13.

Copyright © 2021 Massachusetts Medical Society.